

PROJECT REPORT

SMART STUDY TIME RECOMMENDER USING AI

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PROBLEM STATEMENT

Students often struggle with deciding how much time to allocate to different subjects while preparing for exams. In most cases, study time is distributed randomly or based on guesswork rather than structured planning. This leads to inefficient preparation, where important or difficult subjects may not receive enough attention.

This project addresses this problem by creating a system that recommends how many hours a student should study for each subject based on multiple real world factors.

WHY THIS PROBLEM MATTERS

Effective time management is one of the biggest challenges faced by students. Poor allocation of study time often results in:

- incomplete syllabus coverage
- increased stress before exams
- lower academic performance

A system that can guide students in planning their study time can significantly improve their efficiency and outcomes. Since this problem is directly relevant to everyday student life, it makes the solution both practical and meaningful.

APPROACH TO SOLVING THE PROBLEM

To solve this problem, an AI-based approach was used.

The system takes input parameters such as:

- subject difficulty
- preparation level

- importance of the subject
- number of days left before the exam

Using these inputs, the system predicts the number of study hours required.

A Decision Tree Regressor was used as the machine learning model because it is simple, interpretable, and effective for small datasets. The model was trained on a manually created dataset that represents realistic study scenarios.

In addition to machine learning, a rule-based adjustment system was implemented to refine predictions. For example:

- if the exam is very close, study time is increased
- if preparation is already high, study time is reduced

The system supports multiple subjects and generates a final organized study plan by sorting subjects based on required study time.

KEY DECISIONS MADE

Several important decisions were made during the development of this project:

- Using a Decision Tree model: Chosen for simplicity and interpretability
- Creating a custom dataset: Since real data was not available, a small dataset was designed manually
- Combining ML with rule-based logic: This helped to ensure predictions were more realistic
- Supporting multiple subjects: This made the system more practical and useful as students often have multiple subjects

- **Sorting output:** Helps users prioritize subjects effectively

These decisions helped balance simplicity, accuracy, and usability

IMPLEMENTATION OVERVIEW

The project was implemented using Python.

Libraries used:

- Pandas (for data handling)
- Scikit-learn (for machine learning)

The program follows this flow:

1. Take number of subjects as input
2. Collect details for each subject
3. Predict study hours using the trained model
4. Apply rule-based adjustments
5. Assign priority levels and advice
6. Display a structured and sorted study plan

CHALLENGES ENCOUNTERED

During the development of the project, several challenges were encountered:

- Designing a realistic dataset without real-world data
- Ensuring predictions were not too high or too low
- Balancing machine learning with rule-based logic
- Handling multiple subject inputs efficiently
- Making the output clear and structured

To address these challenges, multiple rounds of testing and refinement were carried out.

Initially, the model was trained on a small manually created dataset. However, early testing revealed that some predictions were unrealistic, especially in extreme cases such as very high difficulty with very low preparation. To improve this, the dataset was adjusted by adding more balanced and varied data points to better represent real study scenarios.

The model was then tested with different combinations of inputs, including edge cases such as:

- very few days left before the exam
- very high preparation levels
- low-importance subjects

These tests helped identify inconsistencies in predictions. To correct this, a rule-based adjustment layer was introduced. For example, study time was increased when the exam was very close and reduced when preparation was already high. This significantly improved the practical accuracy of the results.

Another challenge was handling multiple subjects. Initial versions worked only for a single subject, so the program was redesigned to loop through multiple inputs and store results in a structured format. Additional testing ensured that the program correctly handled multiple entries without errors.

Finally, improvements were made to the output format by organizing results into a sorted table, prioritizing subjects based on required study time. This made the output more readable and useful for real-world application.

Overall, iterative testing and gradual improvements helped refine both the model and the program, resulting in a more reliable and user-friendly system.

WHAT I LEARNED

Through this project, I gained a deeper understanding of:

- how supervised learning works
- how decision tree models make predictions
- how input features affect output predictions
- how to combine machine learning with logical rules
- how to structure a complete AI-based project

This project also improved my problem-solving and programming skills.

CONCLUSION

The Smart Study Time Recommender successfully demonstrates how artificial intelligence can be applied to solve a real-world problem. By analyzing key factors, the system provides structured and meaningful study recommendations.

The project highlights the importance of combining machine learning with logical reasoning to create practical solutions. It can serve as a useful tool for students to improve their study efficiency and time management.

FUTURE IMPROVEMENTS

The project can be further improved by:

- using real student datasets
- developing a graphical user interface (GUI)
- adding daily timetable generation
- deploying it as a web or mobile application